Applications, Advantages & Disadvantages — All 4 Experiments

These are important topics for your **lab viva**. For each experiment, remember why we use it, where it is used, and its main advantages and disadvantages.

Experiment 1 — Pulse Code Modulation (PCM)

Why do we use PCM?

PCM is used to convert an **analog signal into digital binary data** so that it can be transmitted, stored, and processed digitally.

For example, your voice is an analog signal. PCM converts it into binary bits (0s and 1s), making digital communication possible.

Analog Voice → Sampling → Quantization → Binary Encoding → Digital Data

Applications

- Digital telephone systems (PSTN)
- Digital audio recording and WAV files
- Voice communication systems
- Digital transmission systems such as T1 and E1

Advantages

1. High noise immunity compared with analog transmission.
2. Digital signals can be regenerated using repeaters.
3. Easy to store and process digitally.
4. Supports Time Division Multiplexing (TDM).

Disadvantages

1. Requires more transmission bandwidth in many applications.
2. Quantization introduces quantization noise.
3. Requires analog-to-digital and digital-to-analog conversion.
4. More bits per sample increase the bit rate.

Viva answer: PCM is used to convert analog signals into digital form for reliable transmission, storage, and processing. Its main advantage is noise immunity, while its main disadvantages are higher bandwidth requirements and quantization noise.

Experiment 6 — Direct Sequence Spread Spectrum (DSSS)

Why do we use DSSS?

DSSS is used to **spread a signal over a wider bandwidth using a pseudo-noise (PN) code**, making communication more resistant to narrowband interference and jamming.

In your experiment, each original data bit is spread into **7 chips** using a PN code.

Binary Data × PN Code → Spread Signal → BPSK Modulation

Applications

- GPS and other satellite navigation systems using PRN spreading codes
- CDMA mobile communication systems
- IEEE 802.11b Wi-Fi (DSSS-based modes)
- Military and secure tactical radio communications

Advantages

1. Good resistance to narrowband interference and jamming.
2. Provides processing gain through spreading and despreading.
3. Different users can share a frequency band using suitable spreading codes (CDMA).
4. Can improve reliability in challenging radio environments.

Disadvantages

1. Requires greater transmission bandwidth.
2. Transmitter and receiver must synchronize their PN codes.
3. Receiver design can be more complex.
4. Can suffer from the near-far problem in multiuser systems.

Viva answer: DSSS is used to spread a signal using a high-rate PN code, improving resistance to interference and jamming. Its main advantages are processing gain and interference resistance, while its disadvantages include increased bandwidth and the need for code synchronization.

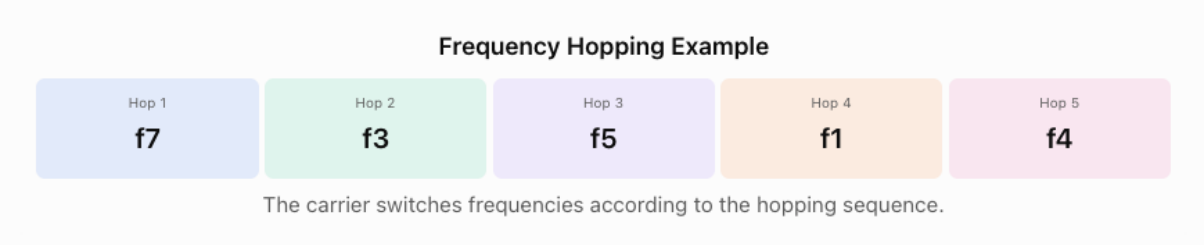
Experiment 7 — Frequency Hopping Spread Spectrum (FHSS)

Why do we use FHSS?

FHSS is used to **change the carrier frequency periodically according to a pseudo-random hopping sequence**, reducing the effect of interference at any one frequency.

Instead of transmitting continuously on one frequency, the transmitter switches among multiple frequencies.

Your experiment uses **7 available hopping frequencies**.



The carrier switches frequencies according to the hopping sequence.

Applications

- Classic Bluetooth wireless communication
- Military radio communication
- Industrial wireless communication systems
- Wireless sensor networks using frequency-hopping protocols

Advantages

1. Good resistance to narrowband interference.
2. Difficult to jam continuously without affecting many hopping frequencies.
3. Reduces the effect of interference concentrated on a single channel.
4. Allows different systems to share spectrum through coordinated hopping.

Disadvantages

1. Requires synchronization between transmitter and receiver.
2. Requires frequency-hopping control and frequency synthesizer hardware.
3. Frequency switching increases system complexity.
4. Interference or collisions can still affect individual hops.

Viva answer: FHSS is used to change the carrier frequency according to a pseudo-random sequence, reducing the impact of narrowband interference and jamming. Its main disadvantages are synchronization requirements and increased hardware complexity.

Experiment 12 — Convolutional Codes

Why do we use convolutional coding?

Convolutional coding is used for **error detection and correction in digital communication systems**.

It adds extra redundant bits to the original message, allowing a suitable receiver decoder to recover information even when some transmitted bits are corrupted.

In your experiment, every input bit produces **2 output bits** using XOR gates and shift registers.

Original Message

1011

Add Two Terminating Zeros

101100

Convolutional Encoder

11 10 00 01 01 11

Code rate = 1/2, constraint length = 3

Applications

- Satellite communication systems
- Deep-space communication
- Wireless and mobile communication systems
- Digital broadcasting
- Error-control systems in noisy communication channels

Advantages

1. Helps detect and correct transmission errors when used with a suitable decoder.
2. Improves communication reliability in noisy channels.
3. Uses memory of previous input bits for effective encoding.
4. Can be decoded using the Viterbi algorithm.

Disadvantages

1. Requires extra bandwidth because redundant bits are transmitted.
2. Encoding and especially decoding introduce additional complexity.
3. Decoding requires computational resources.
4. Can introduce processing delay.

Viva answer: Convolutional coding is used to improve communication reliability by adding redundant bits that help a decoder correct transmission errors. Its main advantage is error-correction capability, while its disadvantages are increased bandwidth and decoding complexity.

Remember: your Experiment 12 program implements **encoding only**. It does not perform error correction or Viterbi decoding.

Final comparison — Memorize this table

Experiment	Main purpose	Biggest advantage	Main disadvantage
1. PCM	Analog-to-digital conversion	Noise immunity	Quantization noise and bandwidth
6. DSSS	Spread signal using PN chips	Interference resistance	Requires wide bandwidth and synchronization
7. FHSS	Spread signal by frequency hopping	Narrowband interference resistance	Hopping and synchronization complexity
12. Convolutional Codes	Add redundancy for error correction	Reliable communication	Extra bits and decoding complexity

Most important viva question: What is the difference between DSSS and FHSS?

DSSS	FHSS
Uses a high-rate PN chip sequence to spread the data signal	Uses a hopping sequence to switch carrier frequencies
Spreading occurs by multiplying data with PN chips	Spreading occurs through successive frequency changes
Receiver needs synchronized PN chips	Receiver needs synchronized hopping sequence
Example: GPS PRN spreading	Example: Classic Bluetooth

One-line memory trick

PCM → Convert Analog to digital **DSSS → Spread** Using PN chips **FHSS → Hop** Between different frequencies **Convolutional Codes → Protect** Add redundancy to enable error correction

For your viva, prioritize remembering the **purpose, one real-life application, two advantages, and two disadvantages** of each experiment. That should help you answer most basic application-oriented questions confidently.